INCH-POUND
MIL-DTL-23919C(SH)
07 August 2012
SUPERSEDING
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03 June 2009

DETAIL SPECIFICATION

REFERENCE CELL, IMPRESSED CURRENT CATHODIC PROTECTION (ICCP) – SILVER-SILVER CHLORIDE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1.1 Scope. This specification covers circular silver-silver chloride flush type reference electrodes used in impressed current cathodic protection systems. The reference electrode provides the control signal to the impressed current system and may be used to monitor the system operation. It includes features of design and fabrication of these electrodes, including alternate components.

1.2 Classification. Silver-silver chloride reference cells are of the following types, grades, classes, and performance levels, as specified (see 6.2):

1.2.1 Type.

a. Type I – Standard Cell

b. Type II – Diver Replaceable Cell

1.2.2 Grade.

a. Grade A – Mesh Reference Cell

b. Grade B – Plate Reference Cell

1.2.3 Class.

a. Class I – Surface Ship

b. Class II – Submarine

1.2.4 Performance levels.

a. Performance A – Standard Performance

b. Performance B – High Performance

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Part or identifying number (PIN). PINs to be used for silver-silver chloride reference cells acquired to this specification are created as follows:

M	**23919**	**-**	**X**	**X**	**X**	**X**
Prefix for Military Specification	Specification Number		Type (see code below)	Grade (see code below)	Class (see code below)	Performance (see code below)

Type code		**Grade code**		**Class code**		**Performance code**	
I	1	A	A	I	1	A	A
II	2	B	B	II	2	B	B
Example: M23919-1A1A – Standard Cell, Mesh Reference Cell, Surface Ship, Standard Performance							

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-C-24231 - Connectors, Plugs, Receptacles, Adapters, Hull Inserts, and Hull Insert Plugs, Pressure-Proof, General Specification for

(Copies of these documents are available online at https://assist.dla.mil/quicksearch/ or https://assist.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E11 - Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

(Copies of this document are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

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SAE INTERNATIONAL

SAE-AMS-P-83461	-	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275 Mdf (135 Mdc)
SAE-AS8660	-	Silicone Compound NATO Code Number S-736

(Copies of these documents are available from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subject to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life-cycle costs.

3.3 Construction.

3.3.1 Silver-silver chloride electrode element.

3.3.1.1 Chemical purity. The silver (Ag) wire used in fabricating the mesh/plate shall not contain less than 99.95 percent silver metal base purity. The silver chloride (AgCl) coating shall also have a metal basis purity of no less than 99.95 percent (see 4.5.7).

3.3.1.2 Cleaning. The silver surface shall be degreased and surface oxide fully reduced (cleaned) by means of a hydrogen flame or equivalent chemical method just prior to application of AgCl. The silver surface shall be bright and shiny once it has been cleaned. The silver shall be handled with stainless steel tongs to prevent contamination.

3.3.1.3 Application of AgCl coating. The AgCl coating shall be applied by either an electroplating or hot dipping process. Sintering shall not be used unless prior approval of the process has been granted by NAVSEA.

3.3.1.4 Surface area. In order to achieve adequate surface area, the Grade A silver element shall have a mesh size of 40 or finer. For a Grade B element, the final surface area of the plate shall be not less than 0.546 inches squared (3.52 centimeters squared) or as approved by NAVSEA (see 4.5.8).

3.3.1.5 Life. The service life of a reference cell shall be 12 years without degradation at input impedance of the reference cell monitoring circuit (see 4.5.9).

3.3.2 Electrical connection. The pin shall be made out of a noble material and plated in 99.95 percent silver metal base purity and shall have a measured resistance of less than 1.0 ohms. Type II reference cells shall be constructed such that the connection to the pin remains dry and free from oxidation after underwater replacement (see 4.5.10).

3.3.3 Assembly.

3.3.3.1 Material. The reference cell assembly shall be made of a non-conductive material to ensure that there is isolation (no shorting or grounding) between the reference cell and the hull, preferably poly-vinyl chloride.

3.3.3.2 Pressure. The reference cell assembly shall be able to withstand the design pressure depending on the class/hull of the specific vessel. After testing both Class I and Class II reference cell assemblies, testing in accordance with 4.5.2 shall be repeated. Reference cells shall show no evidence of mechanical damage, water leakage, or impaired electrical characteristics (see 4.5.6).

3.3.3.3 Shock. The reference cell assembly shall be certified as shock Grade B (see 4.5.11).

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3.3.4 O-rings. O-rings for the hub of the base assembly shall be circular rubber rings conforming to SAE-AMS-P-83461. Sealing grease for installation of the O-rings shall conform to SAE-AS8660.

3.3.5 Fasteners. The fasteners used to attach the assembly to the structure to be protected shall be made from corrosion-resistant material, preferably nylon.

3.4 Performance requirements.

3.4.1 Accuracy/offset. Performance A cells shall have a normalized measurement within ±10 millivolts DC compared to the E244 In-Vivo Metric (IVM) standard reference cell, or equivalent standard cell. Performance B cells shall have a normalized measurement within ±4 millivolts DC as compared to the model E244 IVM standard reference cell, or equivalent cell. Additionally, the difference between the maximum and minimum potential values of all Performance B cells being tested shall not exceed 2 millivolts (see 4.5.2).

3.4.2 Additional requirements for performance B cells.

3.4.2.1 Stability and static drift. For Performance B reference cells, the maximum change between any two reference cell elements shall not exceed 0.6 millivolt DC under constant environmental conditions over 14 days (see 4.5.3).

3.4.2.2 Temperature linearity. For Performance B reference cells, the maximum absolute difference in slope between any two reference cell elements shall not exceed 0.035 millivolt DC/°F when tested at temperatures from 40 to 90 °F over 336 hours (see 4.5.4).

3.4.2.3 Water resistivity linearity. For Performance B reference cells, the maximum absolute difference in slope when plotted on a semi-log graph between any two reference cell elements shall not exceed 0.03 millivolt DC/log(ohm-centimeter) when tested at water resistivities between 18 ohm-centimeters and 288 ohm-centimeters (see 4.5.5).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article sample is required (see 3.1), inspection shall be performed on three sample reference cells of each style of configuration. This inspection shall include all tests of 4.5.

4.3 Conformance inspection. Conformance inspection shall include the tests of 4.5.1 to 4.5.5 and 4.5.7 to 4.5.10.

4.3.1 Lot. For the purpose of sampling, a lot shall consist of those reference cells of each specified classification from the same production run offered for delivery at one time. However, this lot shall not exceed 200 prepared reference cells.

4.3.2 Sampling for examination. From each lot specified in 4.3.1, a minimum of three cells shall be tested, including one of any unique configuration (see table I).

TABLE I. Sampling for examination.

Lot size	Sample size
3-25	3
26-50	5
51-90	6
91-200	7

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4.3.3 Sampling for testing. Sampling for the purpose of testing shall be conducted as dictated by the individual test procedures of 4.5.

4.3.4 Rejection criteria. If a sample fails to pass any of the first article or conformance inspections, the lot shall be considered rejected.

4.4 Summary of requirements and tests. Table II summarizes the requirements and corresponding tests (with reference sections) for verification of satisfying each requirement.

TABLE II. Summary of requirements and tests.

Requirement paragraph	Requirement	First article	Conformance	Test paragraph
3.3.1	Silver-silver chloride electrode element	X	X	4.5.1 and 4.5.7 to 4.5.9
3.3.2	Electrical connection	X	X	4.5.10
3.3.3	Assembly	X	--	4.5.6, 4.5.11
3.4	Performance requirements	X	X	4.5.2 to 4.5.5

NOTE:
1. Performance B reference cells shall pass performance requirements (see 3.4) for both conformance and first article testing. Performance A reference cells shall pass performance requirements (see 3.4) for accuracy and offset testing only.

4.5 Test procedures.

4.5.1 Visual inspection. A non-destructive, visual inspection of the cable, assembly (if applicable), and reference element shall be performed. Upon completion of testing, if necessary, a destructive examination shall be performed to inspect the integrity of the cell element.

4.5.1.1 Cable. The cable shall show no signs of damage or oxidation.

4.5.1.2 Assembly. The assembly shall show no signs of damage. All parts shall be accounted for and in working order, ready to install.

4.5.1.3 Reference element. The reference cell element shall have a surface void of cracks or foreign matter.

4.5.2 Accuracy/offset/stability testing.

4.5.2.1 Calibrated standard reference cell. A NAVSEA-approved standard cell that has been correctly prepared for immersion service shall be used to test product form cells, both Performance A and B. Standard cells used shall meet the following criteria:

a. The sintered silver-silver chloride cell shall have a minimum surface area of 0.0254 square inches (0.164 square centimeters). For a circular cell this would be a 0.18-inch (4.5-millimeter) diameter.

b. The element shall have a lead wire connected that is soldered to the element and fully potted to prevent salt water intrusion into the connection.

c. A reference cell standard tank with a minimum volume of 1.06 gallon (4 liters) shall be provided to accommodate the standard reference cells while conducting the product form testing.

d. The standard tank shall consist of a separate tank from the Performance A or Performance B cell test tank and shall contain 0.6 N (normal) sodium chloride (NaCl) solution.

e. The resistivity of the solution in the standard cell tank shall be 18 ohm-centimeters ±1 ohm-centimeter.

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f. Resistivity shall be checked regularly during all product form testing (minimum once weekly) and distilled water shall be added, as necessary, to maintain the required resistivity.

g. A minimum of 10 standard cells shall be placed in the standard tank for 96 hours at room temperature to stabilize.

h. All air bubbles shall be removed from the cell surface after immersion.

i. DC voltage measurement shall be conducted at the end of the stabilization period to determine the voltage (potential) difference between 9 of the cells using the 10th cell as the base cell to measure against.

j. Three IVM, or equivalent, cells that measure within ±1 millivolt DC of each other versus the base cell shall be identified to be used for further testing as standard cells.

k. One of the three cells, called the base standard, shall be used to compare all product form cells.

l. Other cells shall be maintained as spares in the event one of the standard cells fail.

4.5.2.2 Base standard reference cell failure. If the base standard IVM cell drifts more than ±1 millivolt DC from the initial differential reading between any of the other standard IVM cells, the base standard will be replaced by one of the other two Ag/AgCl standard cells as identified in 4.5.2.1.

4.5.2.3 Offset comparison test. The product form Performance A or Performance B cells shall be immersed in a test tank of sufficient size to contain a minimum volume of water of 0.65 cubic feet (18.4 liters) per reference cell to be tested and be of sufficient proportion to immerse the reference cells while also meeting the following criteria:

a. Two inches (5.08 centimeters) of water below the lower cells shall be provided.

b. Three inches (7.62 centimeters) of water above the uppermost cells shall be provided.

c. Sufficient space shall be provided to accommodate the minimum bend radius of the reference cell cable between the reference cell and the tank wall, with the element face in a vertical position.

d. A minimum of 8 inches (20.32 centimeters) between or in front of each reference cell shall be provided.

e. The transfer pump shall be capable of transferring the seawater to the test tank from the acclimation or mixing tank in less than 10 minutes.

f. The standard tank and the test tank shall be electrically connected via a salt bridge containing 0.6 N NaCl solution (equivalent seawater conductivity).

g. The product form cells shall be allowed to stabilize for up to 2 weeks prior to environmental testing.

h. The test tank electrolyte shall be changed out to a new electrolyte on a daily basis during the stabilization period.

i. After stabilization, the differential DC voltage (potential) between the Performance cells and the standard shall be measured.

j. The potential measurements shall be normalized to 68 °F (20 °C) per the following equation:

$$E_{normalized} = E_{measured} + 0.33 \times \left(20 - T_{@\,E_{measured}}\right)$$

where $T_{@Emeasured}$ is measured in Celsius.

4.5.3 Stability and static drift. Static drift is defined as the inherent deviation of potential versus standard from the initial value over a period of time in which there have been no environmental changes.

$$StaticDrift_n = E_{final} - E_{initial}$$

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4.5.4 Temperature linearity testing.

4.5.4.1 Testing setup. The test tank shall be filled with seawater. Seawater shall not be changed for the duration of temperature cycling. The temperature bath or heating/cooling element shall be set to a temperature that will increase the temperature in the test tank to a minimum of 90 °F (32.2 °C). Temperature shall not exceed 93 °F (33.9 °C). When temperature has reached a minimum of 90 °F (32.2 °C), the temperature control element shall be reversed to begin cooling. The temperature shall be cooled until it reaches a point below 40 °F (4.5 °C). When 40 °F (4.5 °C) has been reached, the temperature of the seawater in the test tank shall be brought back up to the initial starting temperature. The maximum temperature change in a 1-hour period shall not exceed 2 °F (1.1 °C).

4.5.4.2 Data analysis. The DC potential readings between the standard Ag/AgCl cell and each of the product form cells shall be determined every 5 minutes. The potential readings versus temperature shall be graphed. Linear regression line shall be calculated from graphed data. The R-squared value (with minimum of 20 data points) for a linear fit shall meet or exceed a value of 0.9800. While the slope of the line is dependent on the endpoints, the R-squared values will indicate any deviations from linearity between these two points (see figure 1).

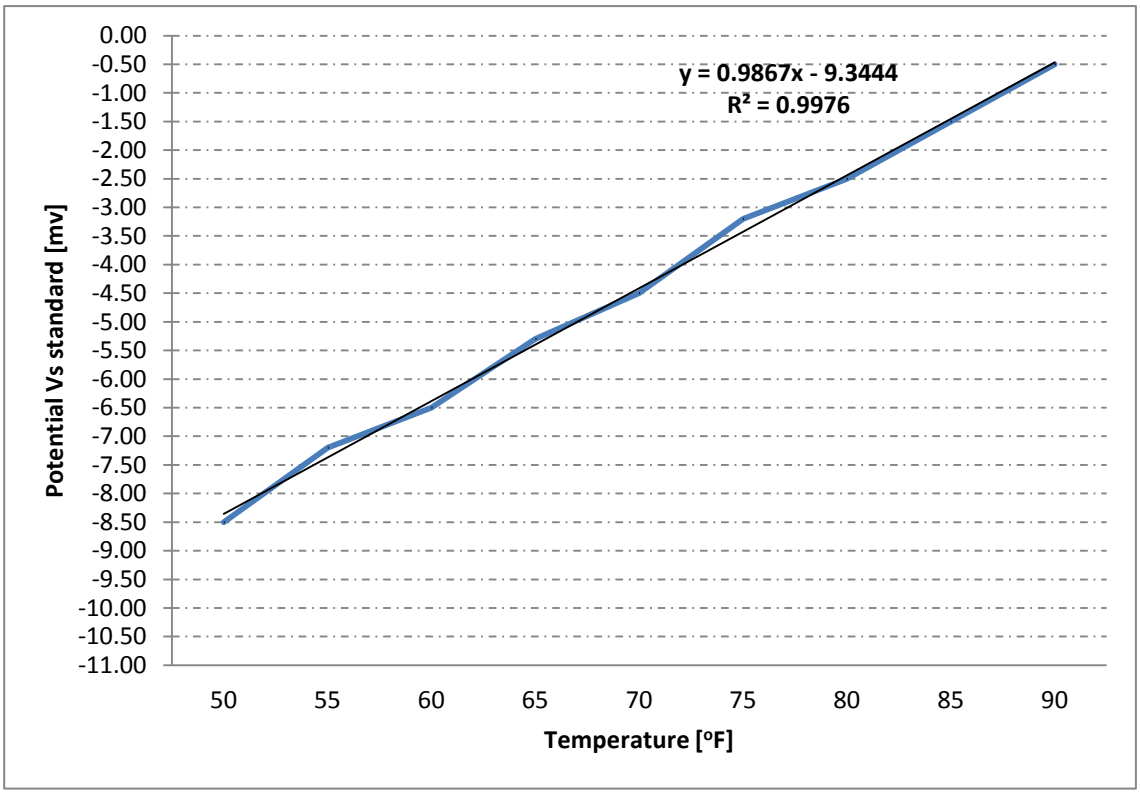


FIGURE 1. Acceptable deviation from linearity during temperature testing.

4.5.5 Water resistivity linearity.

4.5.5.1 Testing setup. The reference electrodes shall be cycled through seawater solutions of varying resistivities. Target resistivities are 18±1, 36±2, 72±5, 144±10, and 288±20 ohm-centimeters. Solutions of varying amounts of seawater and fresh water shall be mixed to achieve the targeted resistivities.

4.5.5.2 Data analysis. The DC potential readings between the standard Ag/AgCl cell and each of the product form cells shall be determined and plotted for each resistivity level. This value shall be measured at the same time index following the water change for each test solution resistivity. This time index shall not be less than 24 hours. Potential readings versus resistivity shall be graphed on semi-log graph. Linear regression line shall be calculated from graphed data. R-squared value shall meet criteria specified in 4.5.4.2. Any deviation from linearity within the upper and lower limits of the test shall be immediate cause for rejection of the reference element.

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4.5.6 Pressure. Each sample electrode shall be mounted in a pressure test fixture so that the face and mating surface of the reference cell assembly shall be covered.

4.5.6.1 Class I reference cells. A pressure of 100±10 pounds/inches squared (70300±7030 kilograms/meters squared) shall be applied to the base, electrode, and pin assembly prior to assembly of the holder (see figure 2). The tube inserted over the electrode hub and O-rings shall be filled with a colored, translucent liquid containing a wetting agent (i.e., Windex® or equal). The pressure shall be maintained for 2 minutes. No bubbles shall appear. Additionally, no deformation shall be present on the reference electrode or assembly.

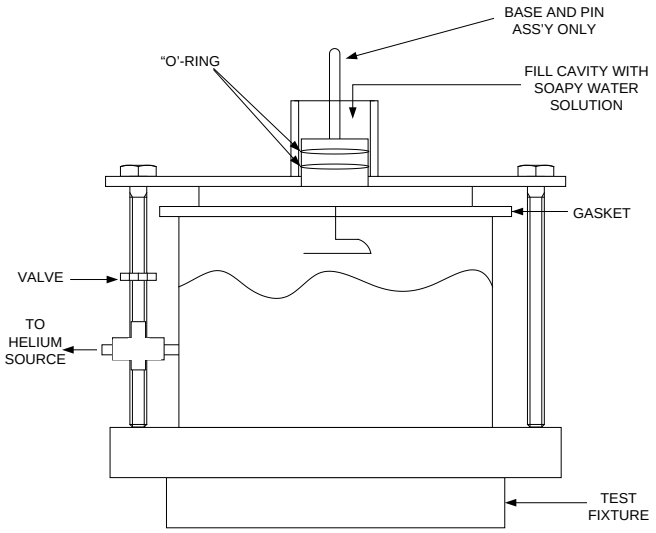


FIGURE 2. Sample pressure test setup for class I reference cells.

4.5.6.2 Class II reference cells. Reference cell pressures are governed by ship specific criteria pertaining to the test depth of the vessel. Reference cell testing shall meet the requirements of MIL-C-24231, Type II, with the exception of insulation resistance, continuity, hardness, and dye-penetrant.

4.5.7 Chemical purity. Proper assessment of Ag and AgCl purity shall be conducted. Certification of this assessment shall be provided.

4.5.8 Surface area. Surface area of a plate style reference cell shall be measured using a calibrated laboratory ruler. Proper surface area of a mesh style reference cell shall be a mesh size of 40 or finer as illustrated in ASTM E11.

4.5.9 Service life. When specified (see 6.2), certification of uniform thickness and longevity (life) calculations shall be provided to support service life requirements. The life shall be based on the weight of AgCl used in the reference cell. Ensure that the minimum amount of AgCl on cell surface is no less than the value given by:

$$Weight_{AgCl}(mg) = \frac{3 \times LeakageCurrent(A) \times ServiceLife(\sec)}{1.49}$$

Where: the leakage current is given as the hull polarization value (e.g., 0.85 volt) divided by the input impedance of the potential measuring circuit (ohms). Note that the amount of AgCl on the reference cell face may typically be more than this value in order to ensure stable readings. The calculated value is only a minimum.

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4.5.10 Underwater replacement. Test will require one reference cell assembly and one additional element. The manufactured assembly shall be placed with original element in natural seawater for a period of 2 weeks. Reference cell voltage versus a standard cell (see 4.5.2.1) shall be measured. After 2 weeks of exposure, the reference element shall be replaced with a new element while underwater without removing the assembly from the test tank. Voltage shall be measured for an additional 2 weeks. At the end of this measurement period, the voltage difference measured shall be within ±10 millivolts of the Performance A elements. Performance B elements shall meet the requirements of 3.4.1.

4.5.11 Shock testing. Shock testing shall be accomplished in accordance with MIL-S-901. When specified (see 6.2), certification of test results shall be provided.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The reference electrodes are primarily intended for use in corrosion prevention in seawater of ship hulls, structures, sea chests, sonar domes, and similar marine structures. The reference electrodes supply the central signal to the automatic control impressed current cathodic protection system.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, grade, class, and performance level (see 1.2).

c. When first article is required (see 3.1).

d. Verification data (see 4.1).

e. Certification of thickness and longevity (life) calculations should be provided to support service life requirements (see 4.5.9).

f. Certification of shock test results should be provided (see 4.5.11).

g. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Control

Corrosion

Electrode

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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